

CE 451: Irrigation Engineering and Hydraulic Structures

Course Code	CE 451
Course Name	Irrigation Engineering and Hydraulic Structures
L-T-P-C	3-0-0-3
Prerequisites	CE 303
Intended for	B.Tech. (Civil Engineering)
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

1. **Crop water requirement:** Scope of irrigation engineering, Irrigation requirements in India, Soil moisture and plant growth, Crop water requirement, Evapotranspiration, Duty, Delta, irrigation scheduling, methods and efficiency, irrigation water quality. (6 Lectures)
2. **Design of Irrigation:** Channel Alignment; canal capacity; losses; FSL of the canal; design of canal in alluvial soil and nonalluvial soils; Kennedy's silt theory; Lacey's regime theory; balancing depth; use of Garrets diagrams and Lacey's Regime diagrams; the lining of irrigation channels; design of lined canal drainage behind lining; Water logging: Causes, Measures: surface and sub-surface drains, land reclamation. (8 Lectures)
3. **Diversion head works:** Introduction; the layout of diversion headwork and its component; Selection of site, Khosla's theory and concept of flow net; safe exit gradient; hydraulic design of weir on Bligh's theory; and design of modern barrage on Khosla's theory; Necessity & functioning of silt excluder & silt extractor. (8 Lectures)
4. **Cross drainage structures:** Types; selection of the suitable type of Cross drainage works; aqueduct and Syphon aqueduct; determination of maximum flood discharge and waterway for drain, fluming of the canal; uplift pressure on the underside of barrel roof and at the floor of the culvert; design of bank connections. (6 Lectures)
5. **Reservoir and planning of dam reservoirs:** Types of dams; selection of the type of dam; capacity elevation and area elevation curves; design of reservoir capacity; Rule curves and operating tables; sedimentation of the reservoir; Gravity dams: modes of failure, stability analysis, construction, joints; Earthen dams: types, modes of failure, design criteria, seepage analysis, and control. (8 Lectures)
6. **Spillway energy dissipators and Spillway gates:** Location of a Spillway; design criteria; controlled and uncontrolled Spillways; Ogee, Chute, Side Channel, Shaft, Syphon Spillways; Energy dissipation; Stilling + basins; Crest gates. (6 Lectures)

Text Books

1. S.K. Garg, **Irrigation Engineering and Hydraulic Structures**, Khanna Publications.
2. B.C. Punmia, **Irrigation and Water Power Engineering**, Laxmi Publication.

References

1. Viessmen, Jr. & Lewis, **Introduction to Hydrology**, PHI Learning Private Ltd.
2. Larry W. Mays, **Water Resources Engineering**, Wiley Publications.